

Application Design and Development (ADaD)

- *The Fellowship of the Code*

Phase 2: Application Design

Deciding

Spring 2026, by Stefan Nafra



0. Agenda

1. Feedback
2. Recapitulation
3. Theory
4. Practice
5. Recapitulation
6. Feedback

Feedback

1. Phase Review



1. Phase Review

*„Despair is only for those who
see the end beyond all doubt.
We do not.“*

- Gandalf



Recapitulation

2. Phase Recapitulation



2. Phase Recapitulation

1. Roadmap
2. Flashback
3. Expectation Management
4. Feedback Management
5. Practice: TFC: Situation & Intent
6. Community Challenge (Meta Game)

2.1. Roadmap



2.2. Flashback

CONTENTS

- Development Basics
 - Introductions
 - Course outline
 - Software concepts
 - Roles & responsibilities
 - Lifecycle

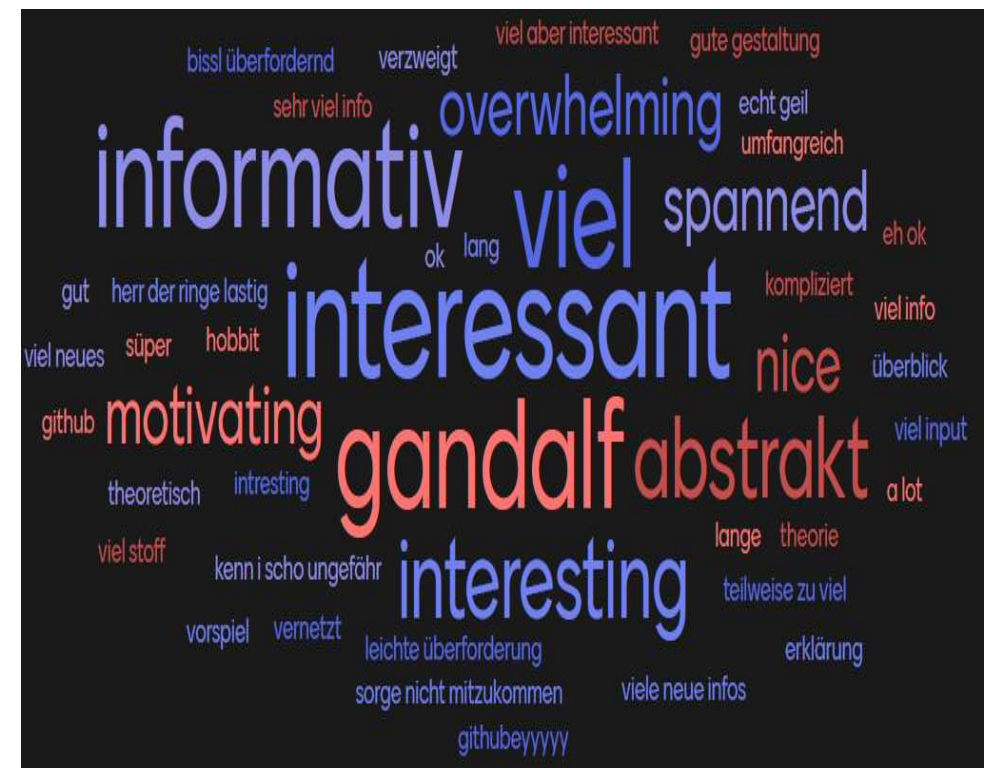
TAKEAWAYS

- Course design & structure
- No matter the role: Someone must own decisions, explain why and accept constraints
- High-level overview of the application design and development activities
 - Planning, analysis, design, implementation, testing, deployment and maintenance

- [illegible]

2.4. Feedback Management

- Positive critics: Thanks <3
- Mixed critics:
 - “viel Input / info”
 - “umfangreich / lang / überziehen”
 - “theoretisch / abstrakt”
 - „viel Text auf Folien“
 - “Sorge” / „herausfordernd“
 - “Herr der Ringe lastig”



2.5. Practice: TFC: Situation & Intent

■ Conceptual foundation practice

- Average scoring: 4.23 / 5 points
- Lessons learned:
 - **Read instructions:** Do not waste valuable points by missing required details (e.g., *naming, linking, missing titles, ...*)
 - **Focus:** A good thing, but it might need an explanation (e.g., *where are the others in your story?*)
 - **The big picture:** This assignment was intended as an outline for the entire application (TFC), so it should also be considered in the context of the entire journey
 - **Value:** Avoid being too generic (*if possible, define an USP*)
 - **Capabilities:** Describe the system's functional power and user-relevant outcomes, independent of specific UI elements, technologies, or implementation details (e.g., *“What can the system do for users?”* and not *“Which features or technology does it use?”*)

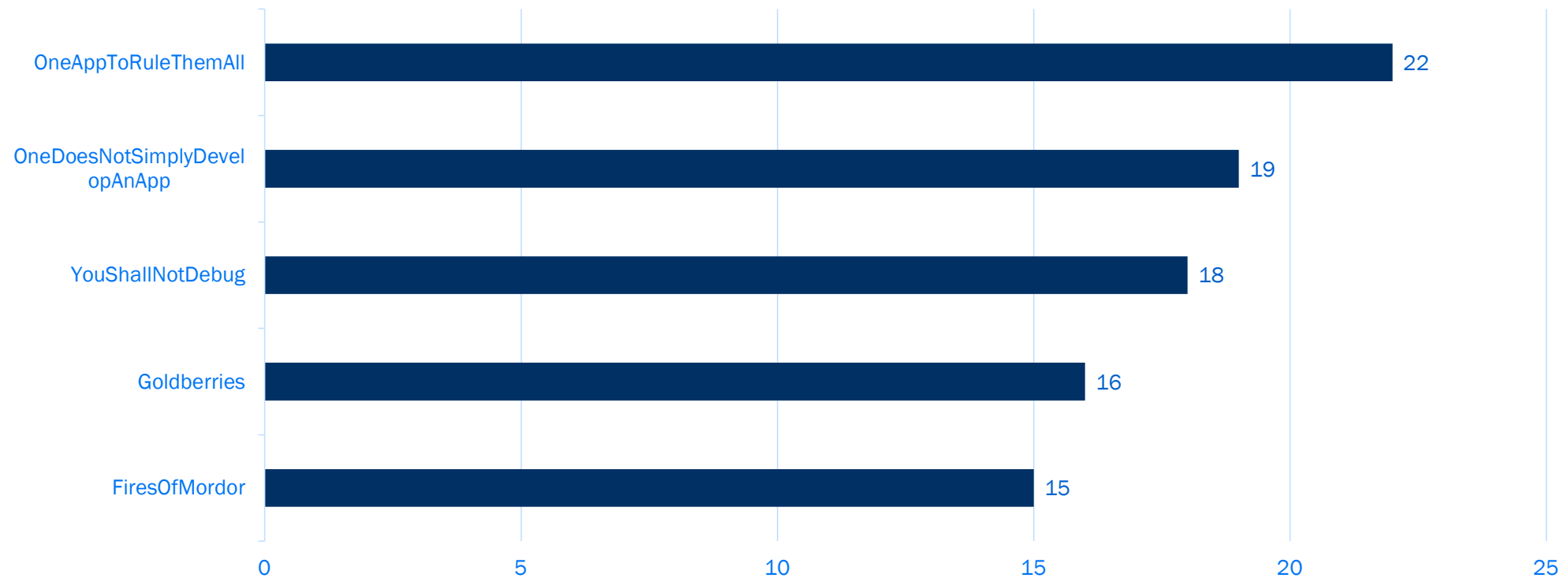
2.6. Community Challenge (Meta Game)

- Your activity on [Lord of the Tweets](#)
 - Tweets: 9 (19*)
 - Reactions: 48 (96*)
 - Comments: 19 (40*)

* = Last year



2.6.1. Activity Score





Break Time



- *“Gentleman, we do not stop before nightfall”*
- *“What about breakfast?”*
 - *“We already had it.”*
- *“We had one, yes. What about second breakfast?”*

Recommended watching:

<https://www.youtube.com/watch?v=gA8LV37QwxA>

Theory

3. Software Concepts



3.

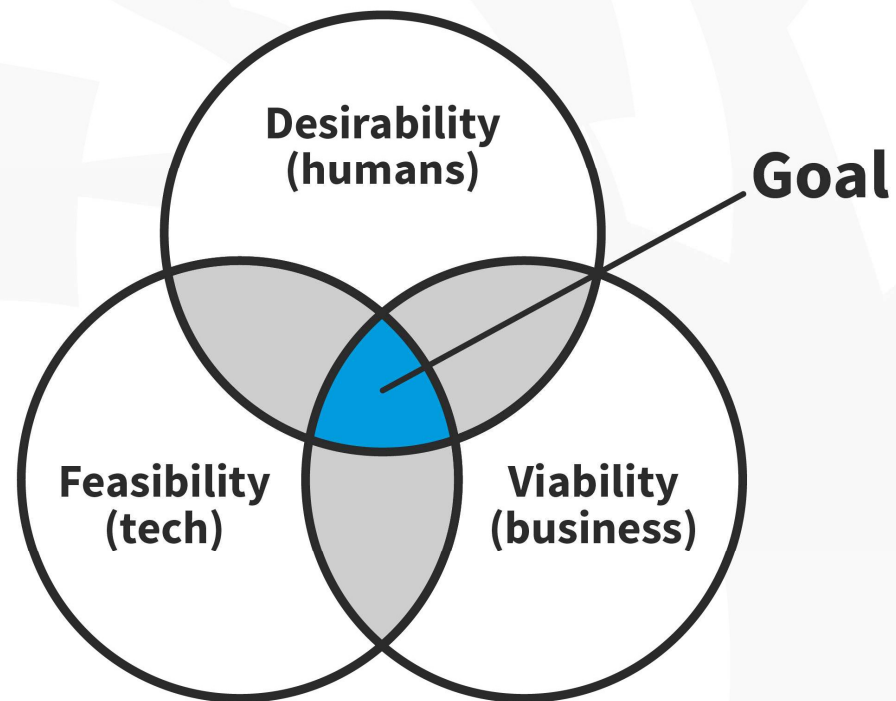
Software Concepts

1. Software Design
2. User Experience & User Interface
3. Decisions
4. Assumptions & Constraints
5. Flows & Designs

3.1. Software Design

- Software is a series of decisions made under uncertainty
- When designing a system, developers and designers must decide:
 - What to model explicitly
 - What to simplify or ignore
 - What the system enforces vs. what it leaves to the user
 - What happens when information is missing or wrong
- **Decisions:**
 - shape user behavior
 - limit future changes
 - encode values into the system
- *Software design is not neutral - logic is not accidental*

3.1.1. Goal



Source: Interaction Design Foundation

3.2. User Experience & User Interface

USER EXPERIENCE

- **Focus:** The overall experience a user has while interacting with a solution
- **Goal:** To ensure the product is usable, enjoyable, and solves the user's problems effectively
- **Key elements:**
 - Researching user needs and behaviors
 - Emphasizing usability, accessibility, and simplicity

USER INTERFACE

- **Focus:** The visual and interactive elements of a product or service
- **Goal:** To make the product aesthetically pleasing and easy to use through well-designed interfaces
- **Key elements:**
 - Designing the layout, color schemes, typography, icons, ...
 - Ensuring visual consistency across the product

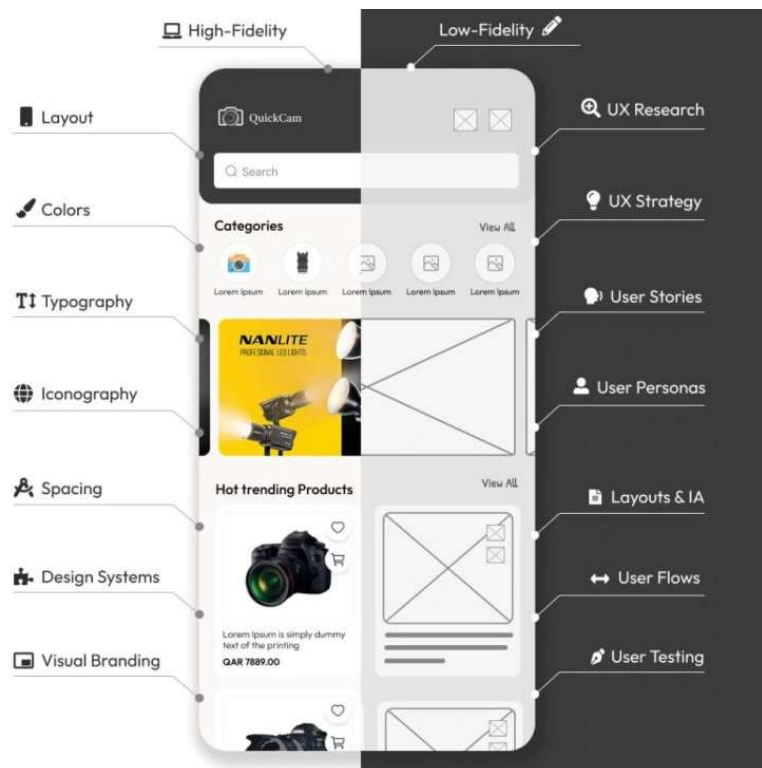
Recommended reading: <https://mouseflow.com/blog/how-to-solve-the-costliest-ux-web-design-friction-points/>

3.2.1. Key Factors



Source: Interaction Design Foundation

3.2.2. Aspects



Source: Intertoons (LinkedIn)

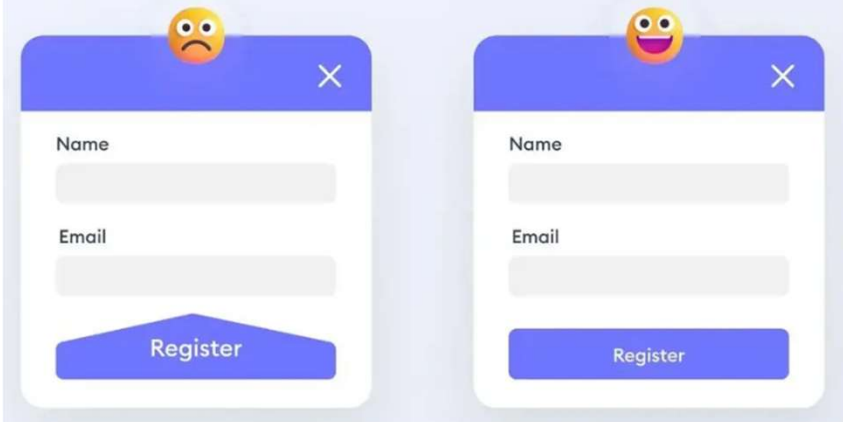
3.2.3. Design Psychology

- Jakob's Law
- Fitt's Law
- Hick's Law
- Miller's Law
- Von Restorff Effect

3.2.3.1. Jakob's Law & Fitt's Law

JAKOB'S LAW

User expect a website to work in similar ways to other website they have visited. Designer should therefore create similar design pattern.

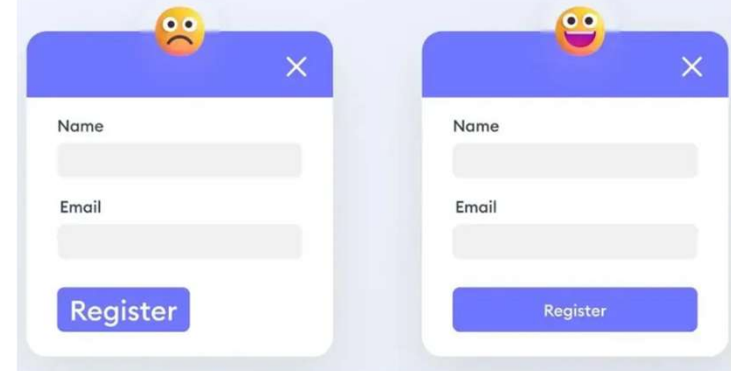


The image shows two side-by-side registration forms. Both forms have a blue header with a close button (X) and a sad face emoji on the left. The forms contain fields for 'Name' and 'Email', and a 'Register' button at the bottom. The left form's 'Register' button is significantly larger and wider than the right form's 'Register' button, which is a standard rectangular button.

Source: Pixelacademy (LinkedIn)

FITT'S LAW

The time taken to acquire a certain target element is a function of its size and distance, so greater the distance to and size of a target element the longer it takes for a user to access them.

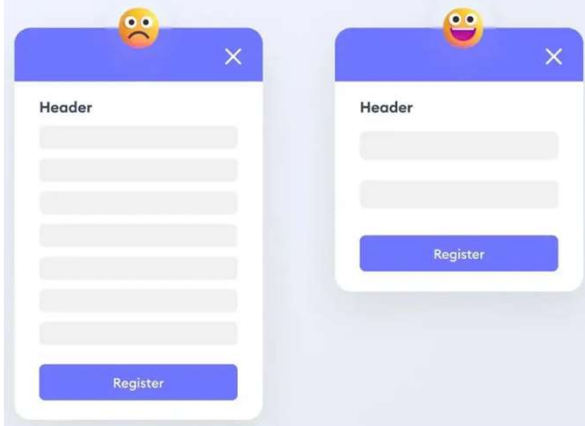


The image shows two side-by-side registration forms. Both forms have a blue header with a close button (X) and a sad face emoji on the left. The forms contain fields for 'Name' and 'Email', and a 'Register' button at the bottom. The left form's 'Register' button is significantly smaller than the right form's 'Register' button, which is a larger rectangular button.

3.2.3.2. Hick's Law & Miller's Law

HICK'S LAW

The time taken to make a certain decision depends on the number of choices available. Present information one at a time according to importance skipping unnecessary items, this improves usability and UX.



Source: Pixelacademy (LinkedIn)

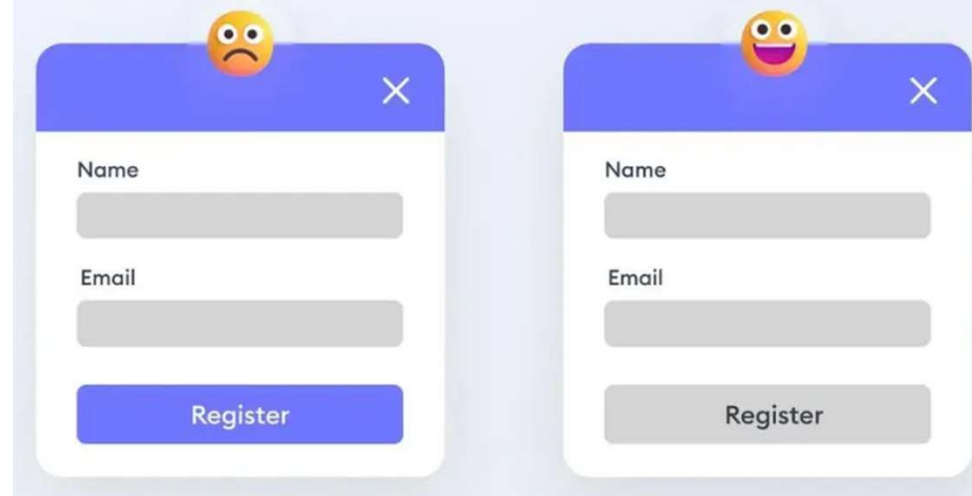
MILLER'S LAW

Users can only keep 7 items in their working memory. This gave birth to the concept of chunking. Present information in little chunks to enable users scan and quickly find what they're looking for.



3.2.3.3. Von Restorff Effect

When similar elements are present, the one that differs from the rest is most likely to be remembered.



The diagram shows two side-by-side registration forms. Both forms have a blue header with a close button (X) and a sad face emoji. The left form has a sad face emoji, while the right form has a happy face emoji. Both forms have input fields for 'Name' and 'Email', and a 'Register' button. The 'Register' button on the left is blue, while the one on the right is grey. This visual contrast highlights the Von Restorff Effect, where the unique element (the happy face and the grey button) is more memorable.

Source: Pixelacademy (LinkedIn)

3.3. Decisions

- Design is the act of deciding how the system behaves under real conditions
- As a designer, you decide:
 - What steps exist in a flow
 - Where choices appear
 - What information is visible, hidden, or delayed
 - How the system reacts to uncertainty
- Decisions:
 - Shape how users think
 - Reduce or increase cognitive load
 - Translate intent into structure
- *Design decisions sit between intent and logic*

Recommended reading: <https://uxdesign.cc/a-framework-to-make-great-design-decisions-ce4a366bf3e1>

3.3.1. Considerations

- Usability
- Maintainability
- Compatibility
- Extensibility
- Security / Privacy
- Performance
- ...

3.4. Assumptions & Constraints

ASSUMPTIONS

- Every design rests on assumptions: Things we take for granted about the users, the environment, or the system
- *“We assume the Fellowship can read maps”*

CONSTRAINTS

- Constraints are the limits that shape how we can implement the design: Time, resources, technology, policies, or unreliable information
- *“The journey must avoid areas of heavy Orc activity; mobile devices are not available”*

Recommended reading: <https://ixdf.org/literature/topics/assumptions>

3.4.1. Assumptions

- User needs
- User goals
- User behavior
- Context of use
- User preferences
- User familiarity
- Culture

3.4.1.1. Benefits & Risks

BENEFITS

- Efficiency
- Creativity
- Hypothesis generation
- Risk management
- Iterative process

RISKS

- Design for the wrong user
- Create irrelevant or unusable features
- Waste time and resources
- Miss out on valuable insights
- Experience difficulty in iteration
- Cause a negative impact on brand perception

3.5. Flows & Designs

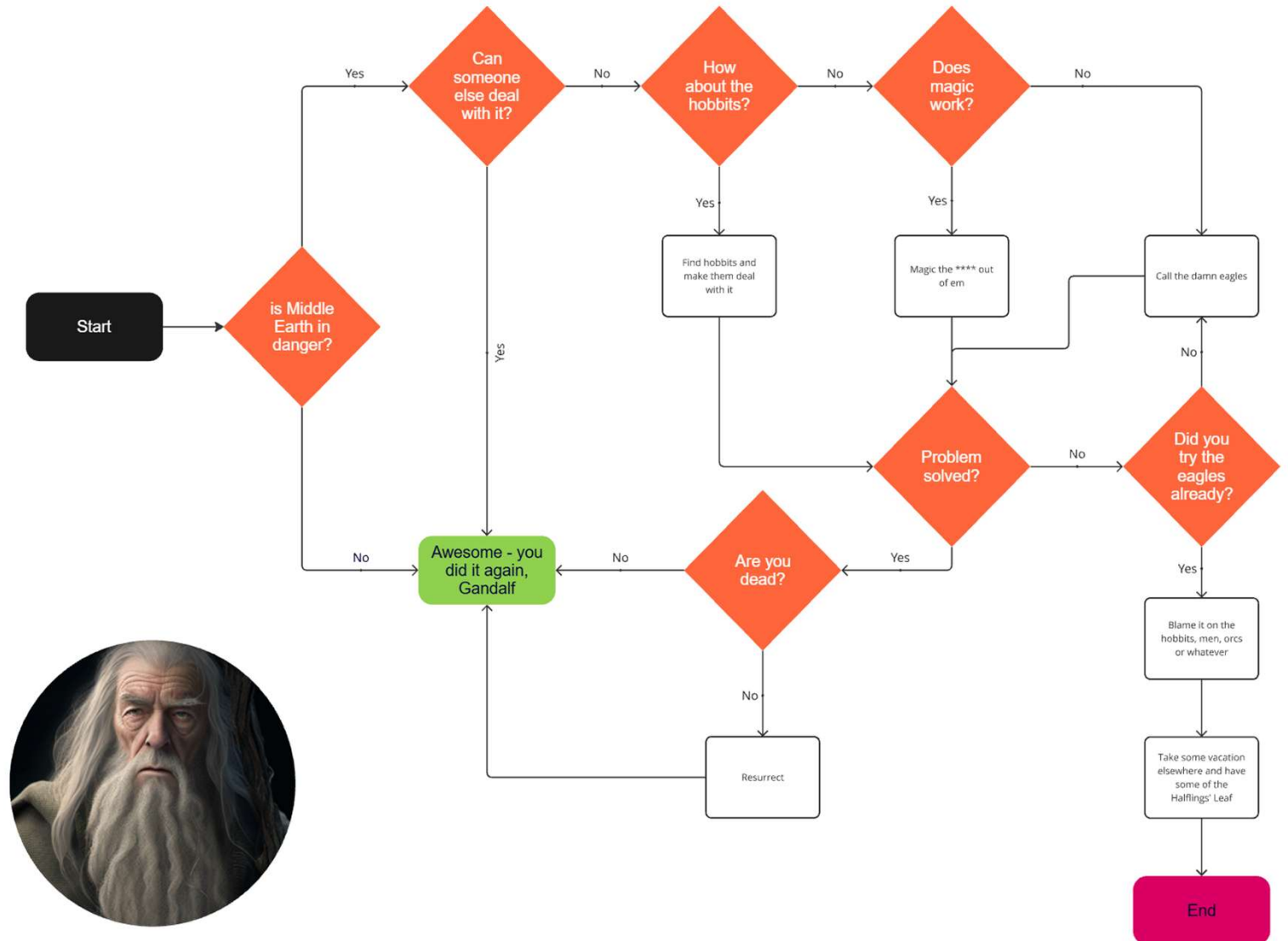
FLOWS

- A flow describes how a user moves through a system to achieve a goal
- Flows are not screens - they are decisions, steps, and transitions
- Good flows:
 - Reduce unnecessary choices
 - Make progress visible
 - Align with the user's mental model

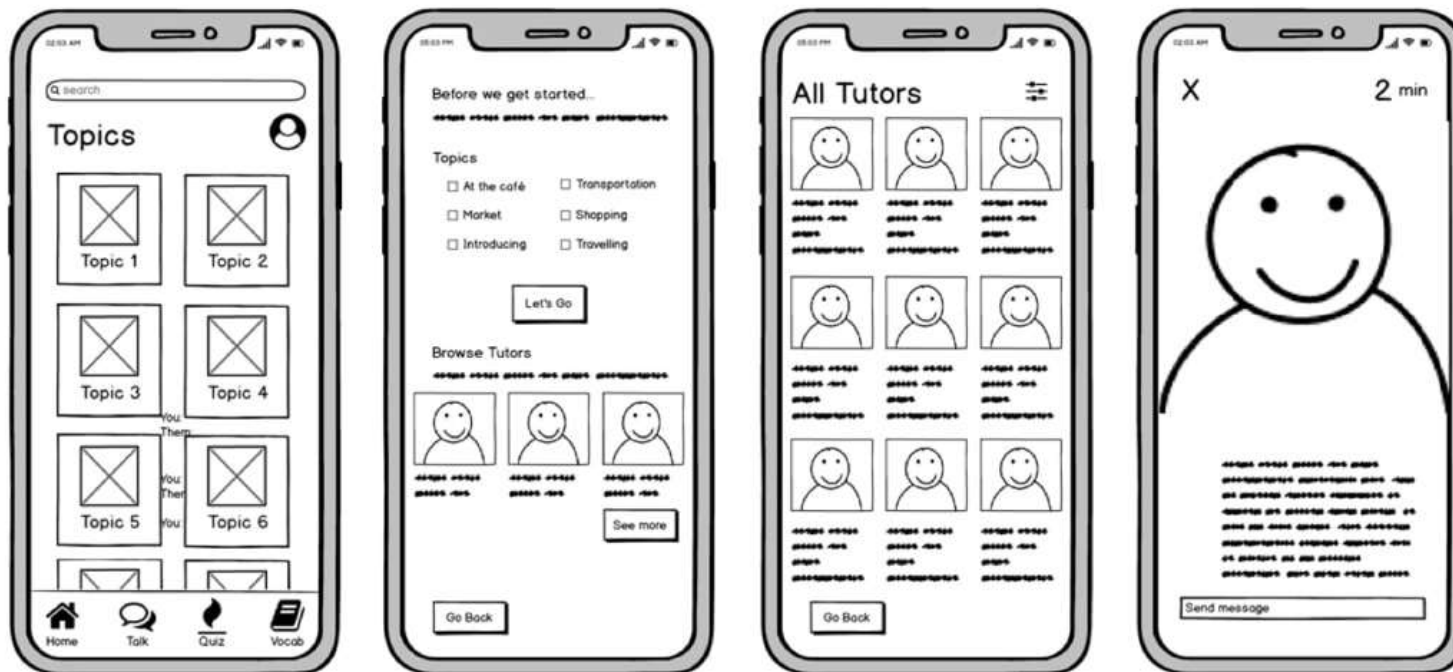
DESIGNS

- A design gives this flow structure, sequence, and emphasis
- Early designs are often expressed through wireframes
 - Visual representations of an interface, stripped down to their essentials
 - Architectural blueprint of your design project
 - Outline the structure and functionality of your product

3.5.1. Flow



3.5.2. Wireframe



Source: <https://careerfoundry.com/en/blog/ux-design/what-is-a-wireframe-guide/>



Break Time



- *“Gentleman, we do not stop before nightfall”*
- *“What about breakfast?”*
 - *“We already had it.”*
- *“We had one, yes. What about second breakfast?”*

Recommended watching:

<https://www.youtube.com/watch?v=gA8LV37QwxA>

Practice

4. Decisions

5. Flows



4.1. Decisions

CONTEXT

- „The Fellowship must decide which path to take next. Information is unreliable. Time is limited.”
- Proposed Design:
 - The system presents three paths as buttons:
 - “Mountain Pass”, “Forest Route”, “River Crossing”
 - Each path shows a short label:
 - “Fast”, “Safe”, “Hidden”
 - When a path is clicked:
 - The system immediately confirms the choice: “Path selected - Proceed”
 - The decision cannot be changed
 - No additional context is shown

QUESTIONS

- What assumptions did the designer make?
- What decision did the designer already make for the user?
- What sense of certainty does this design create?
- How does this design push behavior instead of supporting it?
- What risk does this design introduce into the system?

4.1.1. Answers?

- **What assumptions did the designer make?**
 - The user already understands the risks of each path
 - Labels such as “Fast” or “Safe” are sufficient
 - Speed is more important than comprehension
 - Commitment should be immediate
- **What decision did the designer already make for the user?**
 - There is no option to change your mind
 - High-risk commitment
- **What false sense of certainty does this design create?**
 - The meaning of “safe”
 - There is unreliable information
- **How does this design push behavior instead of supporting it?**
 - No option to compare
 - There are no visible consequences before commitment
- **What risk does this design introduce into the system?**
 - Choice regrets (user)
 - Unclear responsibility

4.1.2. Improvements?

- Add a comparison step
- Show uncertainty indicators
- Delay confirmation
- Allow reconsideration

5. Assumptions & Constraints

ASSUMPTIONS

- *“We assume the hobbits will always remember to update the step count”*
- *“We assume the map data is always accurate”*

CONSTRAINTS

- *“The interface must fit on a single page”*
- *“The system updates only when a button is pressed”*

5.1. Implications

- These assumptions and constraints are
 - arbitrary,
 - disconnected from the actual journey, and
 - ignore realistic uncertainties

5. Flows

SCENARIO

- *The Fellowship must decide how to continue their journey*
- They have two possible paths:
 - Go through Moria (*dangerous*)
 - Go around the mountains (*safer*)
- Their decision depends on:
 - Whether the mountain pass is blocked
 - Whether the Fellowship has weapons

TASK

- Create a simple flowchart that shows:
 - A start and end
 - At least two decision points
 - One final decision for each path
 - Use Mermaid flowchart syntax
 - Keep it simple - no edge cases, no extra logic

Recommended reading: <https://mermaid.ai/open-source/intro/> & <https://mermaid.ai/open-source/syntax/flowchart.html>

Recapitulation

6. Lecture Recapitulation

7. Self-Study Assignments

6.

Lecture

Recapitulation

- 1. Flashback
- 2. Quiz

6.1. Flashback

CONTENTS

- Development Basics
 - Software Design
 - User Experience & User Interface
 - Decisions
 - Assumptions & Constraints
 - Flows & Designs

TAKEAWAYS

- Software design shapes user behavior, limits future changes, and encodes values into the system
- The difference between user experience and user interface
- Decisions shape how users think, reduce or increase cognitive load, and translate intent into structure
- The benefits and risks of assumptions
- The purpose of flows and designs (*wireframes*)

6.2. Quiz

„You shall not pass.“
- Gandalf



Recommended watching:

<https://www.youtube.com/watch?v=3xYXUeSmb-Y>

6.2.1. Software Design: What does it mean to say “software is a series of decisions made under uncertainty”?

- When we design software, we never know everything in advance
- We must decide what to model, what to simplify, what the system enforces, and what it leaves to users - even when information is incomplete or wrong
- => *Design is therefore never neutral or accidental*

6.2.2. UX and UI: How do UX and UI differ, and why do we need both?



- UX focuses on the overall experience - whether the product is useful, usable, and solves real user problems
- UI focuses on the visual and interactive layer - layout, colors, typography, and consistency
- => *A product can look good (UI) but still feel frustrating (UX), so both are necessary*

6.2.3. Decisions: What kinds of things are we deciding when we design a system?

- We decide what steps exist in a flow, where choices appear, what information is visible or hidden, and how the system reacts to uncertainty
- => *These decisions shape how users think and how much effort they need to use the system*

6.2.5. Decisions: Why are design decisions described as “between intent and logic”?

- Intent is what we want to achieve (user goals, business goals)
- Logic is what the system can do technically
- => *Design decisions translate intent into structured behavior that logic can implement*

6.2.6. Assumptions: What is an assumption in design? Can you name one example?

- An assumption is something we take for granted about users, context, or the system
- Example: Assuming users are familiar with maps, understand certain symbols, or have prior knowledge
- => *If assumptions are wrong, the design can fail*

6.2.7. Constraints: What are constraints, and how do they affect design?



- Constraints are limits such as time, resources, technology, policies, or environment
- They shape what is possible and often force trade-offs
- => *Good design works with constraints instead of ignoring them*

6.2.8. Flows & Designs: What is the difference between a flow and a design?

- A flow describes how a user moves through decisions and steps to reach a goal
- A design gives that flow structure and emphasis, often through wireframes that show layout and hierarchy
- => *Flows are about movement; designs are about structure*

7. Self-Study Assignments

#	TITLE	TYPE	DUE DATE	TASKS	TOOLING
5	TFC: Decisions	PRAC	21.04.2026	• Design a user decision flow • Design a static interface	• Mermaid • Any design tool (e.g., Miro) • GitHub & GitHub Codespaces
6	TRDoW: Chapter II - Paths & Choices	PRAC	21.04.2026	• Document & reflect • Share & care • Prepare	• GitHub & GitHub Codespaces

Feedback

8. Lecture Review





8. Lecture Review



*„Despair is only for those who
see the end beyond all doubt.
We do not.“*

- Gandalf